Owner's Manual: Northfen PW-1900E Electric Pressure Washer

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Important safety information

Read and understand every safety message before you set up or operate the Northfen PW-1900E Electric Pressure Washer.

Signal words

Safety messages in this manual use these signal words:

DANGER	A hazardous situation that, if not avoided, will cause death or serious injury.
WARNING	A hazardous situation that, if not avoided, could cause death or serious injury.
NOTICE	A situation that could damage the machine or other property but will not cause injury.

Safety messages for this machine

⚠ WARNING

Injection hazard from high-pressure spray.

The spray can cut skin and inject water and chemicals into the body. An injection can look like a small cut but can cause serious injury or loss of a limb.

Never point the spray gun at yourself, other people or animals.

Never put your hand or fingers in front of the nozzle.

Engage the trigger lock whenever you are not spraying.

If spray pierces the skin, get emergency medical care right away. Tell the doctor it is an injection injury.

⚠ WARNING

Electric shock.

Water and electricity together can cause death or serious injury.

Plug the power cord only into a grounded outlet.

Keep the plug and the GFCI device on the cord out of water.

Test the GFCI device before each use. Do not use the machine if the test fails.

Do not use an extension cord.

⚠ WARNING

Stored pressure.

The hose and spray gun stay pressurized after you stop the machine. Disconnecting a pressurized part can release a burst of water and cause injury.

Relieve pressure before you disconnect the hose, change a nozzle, or service the machine.

Personal protective equipment

Wear this equipment every time you operate the machine:

- Safety glasses or goggles that meet ANSI Z87.1
- Closed-toe, slip-resistant shoes
- Long pants

How the pressure washer works

The PW-1900E takes water from a standard garden hose and uses a pump to raise its pressure to as much as 1,900 psi (131 bar). The spray gun and nozzle turn that pressure into a focused stream that removes dirt, mildew and loose paint.

The path water takes

Water enters the pump through the garden hose inlet. A screen in the inlet catches debris that could damage the pump. The electric motor drives the pump, which pushes the water through the high-pressure hose to the spray gun. The nozzle on the end of the spray wand narrows the stream and sets the spray pattern.

What happens when you release the trigger

When you release the trigger, water has nowhere to go. An *unloader valve* senses the rise in pressure and sends the water back through the pump in a loop. This is called *bypass mode*. On the electric model, the motor also shuts off automatically after a few seconds in bypass mode and restarts when you squeeze the trigger again.

Water that loops in bypass mode heats up. A *thermal relief valve* releases hot water from the pump before it can damage the seals. If you see water dripping from under the pump after a long pause, the valve is working as designed.

Main components

Garden hose inlet	Connects the water supply. Contains the inlet screen.
Pump	Axial cam pump, sealed and maintenance-free.
Power cord with GFCI	A ground fault circuit interrupter on the cord cuts power if it detects current leaking to ground.
High-pressure hose	Carries pressurized water from the pump to the spray gun.
Spray gun and wand	The trigger starts and stops the spray. The trigger lock prevents accidental spraying.
Quick-connect nozzles	Color-coded tips that set the spray angle.

Choosing a nozzle

A narrower spray angle delivers more force to a smaller area. Start with the widest nozzle that does the job, and move to a narrower one only if needed.

Color	Angle	Typical use
Yellow	15°	Stripping loose paint and cleaning concrete.
Green	25°	General cleaning of decks, driveways and patio furniture.
White	40°	Rinsing vehicles, windows and siding.
Black	65°	Applying detergent at low pressure.

Set up and start the pressure washer

Connect the water supply and hoses, purge air from the system, and start the PW-1900E.

You need:

- A garden hose no longer than 50 ft (15 m) and a water supply of at least 1.5 gal/min (5.7 L/min)
- A grounded 120 V outlet on a 15 A circuit
- The personal protective equipment listed in Important safety information on page 3

⚠ WARNING

Electric shock.

Water and electricity together can cause death or serious injury.

Plug the power cord only into a grounded outlet.

Keep the plug and the GFCI device on the cord out of water.

Test the GFCI device before each use. Do not use the machine if the test fails.

Do not use an extension cord.

Place the machine on a level surface outdoors, where spray and runoff cannot reach electrical outlets or open drains.

1. Check that the inlet screen is clean and seated in the garden hose inlet.

   Rinse the screen under running water if you see debris.
2. Connect the garden hose to the garden hose inlet and tighten it by hand.
3. Connect the high-pressure hose to the pump outlet and to the spray gun.

   Pull back on each quick-connect fitting to confirm it is locked.
4. Insert a nozzle into the end of the spray wand.

   For a first test, use the white 40° nozzle. See Choosing a nozzle on page 4.
5. ***NOTICE***

   Pump damage.

   Running the pump without water overheats the seals in seconds.

   Turn on the water supply and purge air from the system before you start the machine.

   Turn on the water supply fully. Release the trigger lock, point the spray gun at the ground, and squeeze the trigger until a steady stream of water flows with no air sputtering.

   Air is purged from the pump and hoses.
6. Plug the power cord into the outlet and press **TEST** and then **RESET** on the GFCI device.

   The GFCI indicator light turns on.
7. Turn the power switch to **ON**.
8. **⚠ WARNING**

   Injection hazard from high-pressure spray.

   The spray can cut skin and inject water and chemicals into the body. An injection can look like a small cut but can cause serious injury or loss of a limb.

	Never point the spray gun at yourself, other people or animals. Never put your hand or fingers in front of the nozzle. Engage the trigger lock whenever you are not spraying. If spray pierces the skin, get emergency medical care right away. Tell the doctor it is an injection injury.

Hold the spray gun with both hands and squeeze the trigger to begin cleaning.

Start with the nozzle at least 24 in. (60 cm) from the surface and move closer slowly. Engage the trigger lock whenever you pause.

The pressure washer is running and ready for use. When you finish, follow Shut down, relieve pressure and store the pressure washer on page 6.

Shut down, relieve pressure and store the pressure washer

Stop the PW-1900E, release trapped pressure, and drain the pump so the machine is safe to handle and ready for storage.

Follow these steps every time you finish using the machine, and before you change a nozzle, clean the inlet screen, or perform any maintenance.

⚠ WARNING	
	Stored pressure. The hose and spray gun stay pressurized after you stop the machine. Disconnecting a pressurized part can release a burst of water and cause injury. Relieve pressure before you disconnect the hose, change a nozzle, or service the machine.

1. Engage the trigger lock on the spray gun.
2. Turn the power switch to **OFF** and unplug the power cord.
3. Turn off the water supply.
4.

⚠ WARNING	
	Injection hazard from high-pressure spray. The spray can cut skin and inject water and chemicals into the body. An injection can look like a small cut but can cause serious injury or loss of a limb. Never point the spray gun at yourself, other people or animals. Never put your hand or fingers in front of the nozzle. Engage the trigger lock whenever you are not spraying. If spray pierces the skin, get emergency medical care right away. Tell the doctor it is an injection injury.

   Point the spray gun at the ground, release the trigger lock, and squeeze the trigger until water stops flowing.

   Pressure is relieved from the hose and spray gun.
5. Engage the trigger lock again.

6. Disconnect the garden hose, the high-pressure hose, and the spray gun.
7. Drain the water from the pump, hoses and spray gun.

   Tip the machine so the pump outlet faces down, and hold each hose vertically until it stops dripping.
8. **NOTICE**

   Freeze damage.

   Water left in the pump can freeze, crack the pump and void the warranty.

   Drain all water and add pump protector before you store the machine where temperatures can fall below 32 °F (0 °C).

   If you will store the machine for more than 30 days or below freezing, add pump protector.

   Attach the pump protector bottle to the garden hose inlet and squeeze until foam comes out of the pump outlet.
9. Store the machine and accessories indoors in a dry place.

The machine is depressurized, drained and protected for storage.

Specifications

Technical data for the Northfen PW-1900E Electric Pressure Washer.

Table 1: Specifications for the PW-1900E

Specification	Value
Maximum pressure	1,900 psi (131 bar)
Flow rate	1.2 gal/min (4.5 L/min)
Power	120 V AC, 60 Hz, 13 A
Pump	Axial cam, sealed
High-pressure hose	25 ft (7.6 m)
Nozzles included	15°, 25°, 40°, 65° (soap)
Maximum inlet water temperature	104 °F (40 °C)
Weight	29 lb (13.2 kg)

Troubleshooting

Find the symptom in the table, then try each solution in order. If the problem continues, contact an authorized service center.

⚠ WARNING

Stored pressure.

The hose and spray gun stay pressurized after you stop the machine. Disconnecting a pressurized part can release a burst of water and cause injury.

	Relieve pressure before you disconnect the hose, change a nozzle, or service the machine.

Before you inspect or service any part, complete Shut down, relieve pressure and store the pressure washer on page 6 through the step that relieves pressure.

Table 2: Common problems and solutions

Problem	Possible cause	Solution
Motor does not start	GFCI device has tripped	Press **RESET** on the GFCI device. If it trips again, stop using the machine and contact service.
	Circuit breaker has tripped	Reset the breaker. Make sure no other devices run on the same circuit.
Low or pulsing pressure	Not enough water supply	Open the water supply fully. Use a hose no longer than 50 ft (15 m) and check it for kinks.
	Air in the system	Stop the machine and purge air as described in Set up and start the pressure washer on page 5.
	Nozzle is worn or partly blocked	Clear the nozzle with the cleaning pin, or replace it.
Water leaks at a hose connection	Connection is loose or O-ring is damaged	Tighten the connection by hand. Replace the O-ring if the leak continues.
Water drips from under the pump	Thermal relief valve is releasing hot water	This is normal after a long pause in bypass mode. Squeeze the trigger more often, or stop the machine during long breaks.
Detergent does not flow	High-pressure nozzle is installed	Install the black 65° soap nozzle. Detergent flows only at low pressure.